

Analyzing Trends in Credit Card Data

Group 1: Nora Fraley, Samhita Kolluri, Tanvi Gadiraju,
Varnika Mukku, Keira Wesley, Sophia Girvin

Introduction

For many, financial literacy is often overlooked or not fully understood, especially as students. This project analyzes the **Credit** dataset from "An Introduction to Statistical Learning (ISL)" by James, Witten, Hastie, and Tibshirani in order to explore relationships amongst credit card users. This data set contains information on ten thousand simulated customers, with 11 quantitative and categorical variables offering different information about these individuals. The categorical variables include individuals' ethnicity, gender, student status, and marital status, while the quantitative variables include individuals' income, credit limit, credit score, number of cards owned, age, education (in years of schooling for example, a high school diploma would mean education = 12), and credit card balance. There is also an ID variable that, while numerical, simply denotes which individual we are looking at and has no real meaning.

Research Questions

- What factors have the greatest influence on an individual's credit limit?
- What relationships exist between students and credit card data?
 - Do students have higher credit scores than non-students?
 - Do non-students have higher credit card utilization than students?
- What is the correlation between a person's credit score and their marital status?
 - Do married individuals have higher credit limits than unmarried individuals?
 - Is there a relationship between marital status and credit score?

Our goal in this study is to find out how different factors contribute to a person's financial success. We want to know the ways in which unchanging values (like a person's demographics) and changing values (like a person's income or student/marital status) affect different aspects of their credit card information and financial status.

Methods

What factors have the greatest influence on an individual's credit limit?

For this question, we first chose income, credit score, number of credit cards owned, and age as our variables to test and plotted each of these variables against credit limit. Then, we calculated the correlation coefficient of each variable separately and decided to further investigate only credit card rating and income through multiple regression. We then calculated each coefficient's p-value and compared them, as well as testing each variable's importance to the model by comparing adjusted R^2 values with/without variables (i.e. if we remove the variable of credit rating but keep income, does R^2 increase?).

What relationships exist between students and credit card data?

To investigate the relationship between students and credit card data, we first filtered the dataset to include only students and created a new variable called utilization, calculated as balance divided by credit limit. We then conducted a two-sample t-test to compare credit ratings between students and

non-students, as well as another two-sample t-test to compare credit card utilization between students and non-students. We hypothesized that students would have lower ratings than non-students but would have higher utilization.

What is the correlation between a person's credit score and their marital status?

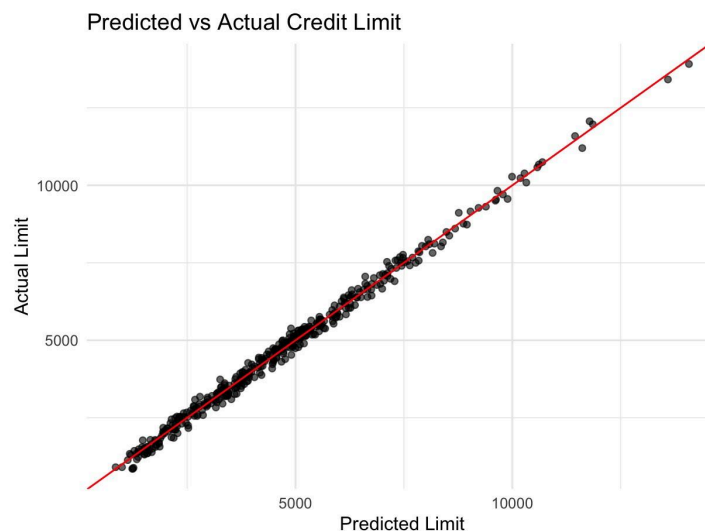
To analyze married and unmarried individuals' credit limits, we first created boxplots to compare the distribution of mean credit limits between married and unmarried people. When we noticed a slight difference, we wanted to investigate whether it was significant. So, we created a bootstrap distribution of the difference in mean credit limit between married and unmarried individuals. Using the infer package, we set credit limit as our response variable and marital status as our explanatory variable so that we could study how credit limit changed based on cardholders' marital status. We generated 10,000 bootstrap samples to calculate the difference in means (married individuals - unmarried individuals). Then, we plotted a histogram of the results and calculated a 95% confidence interval to find the true difference in mean.

For the correlation between marital status and credit score, we first created a violin plot to show the distribution of credit scores for married and unmarried people. Then, we conducted a Chi-square test for independence to investigate whether there is a relationship between marital status and credit score.

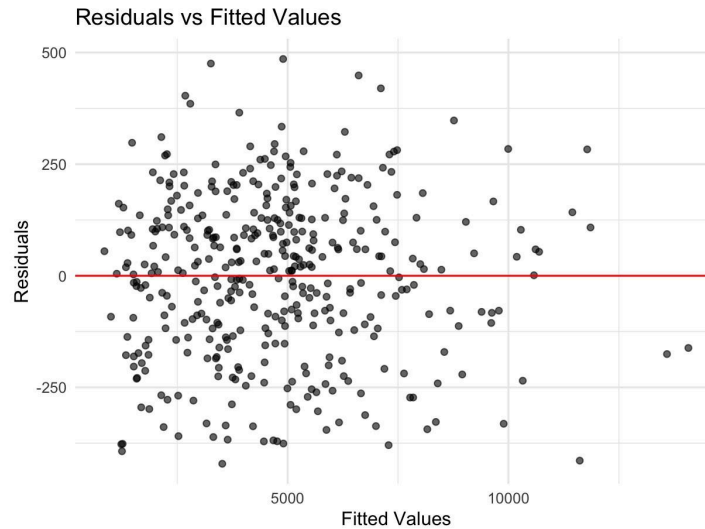
Results

Which variables are most associated with credit limits?

First, by examining individual plots of each variable as compared to credit limits, we deduced that the factors to focus on were credit card rating and income, as they had the highest correlations to credit limit (correlations of 0.997 and 0.792, respectively). While credit card balance also had a high correlation to credit limit (correlation of 0.86), we felt that this was more a result of credit limit rather than a possible predictor for limit, since having a balance on one's credit card is due to exceeding credit limit; the rest of the variables had weak or no correlation to credit limit. We then calculated the multiple regression line and residual plot (shown below). Since the regression line's residuals are normally distributed, this model accurately fits the data.



Figures 1 and 2: Multiple Regression Plot (top) and Residuals vs. Fitted Values (bottom)



Our regression line produced was:

$$L = -532.47 + 14.77R + 0.56I,$$

where L , R , I are credit limit, credit rating, and the individuals' income, respectively. For every unit increase of credit rating, credit limit increased by 14.77, as compared to the per unit increase of income increased limit by 0.56. The p-values of rating and income were $p = 0$ and $p = 1 \times 10^{-89}$, respectively. Both values are much lower than the significance level of $\alpha = 0.05$, thus indicating rating are major predictors of credit card limit. Further, the adjusted R^2 value is 0.9938 for our regression line and a plot of the residuals shows a random distribution, indicating that our model fits the data well. From the regression line, credit rating has the more significant impact on credit limit since its p-value is lower and, when comparing adjusted R^2 values of the regression line with only rating or limit as predictor, we see that rating as a single predictor has a much higher adjusted R^2 value (adjusted R^2 values for each of these models were 0.9938 and 0.6265).

Do students have higher credit utilization as compared to non-students?

When we compared utilization between students and non-students, we calculated an observed difference in means (sample statistic) of 0.0912. This value was plotted alongside our generated null distribution for difference in means and is shown below.

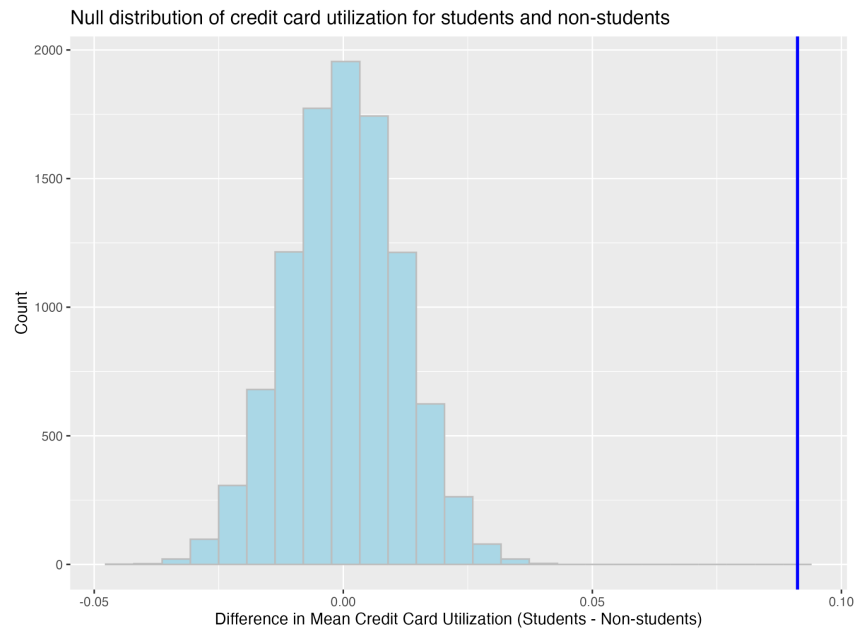


Figure 3: Credit utilization for students and non-students

In order to test this difference, we conducted a two-sample t-test for difference in means using the following hypotheses.

$$H_0 : \mu_{\text{student}} = \mu_{\text{non-student}}$$

$$H_A : \mu_{\text{student}} > \mu_{\text{non-student}}$$

From this test, we calculated a p-value of $p = 0$. Since this value is lower than our significance level of $\alpha = 0.05$, we can conclude that there is convincing evidence that students have higher credit card utilization than non-students.

Do non-students have higher credit scores than students?

Following a similar method to the previous question, we calculated an observed difference in means and plotted the null distribution for difference in means, this time for credit scores. Our observed difference was 1.0444 and our null distribution plot is below. Notably, our difference in means is swapped here, as we believe non-students have less erratic spending habits and thus will have higher credit scores, so our direction for difference was non-students - students.

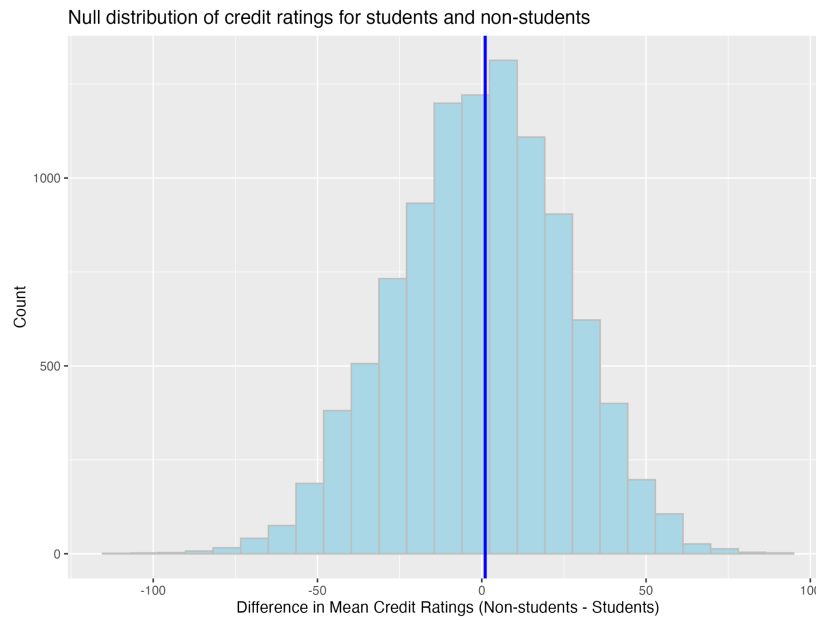


Figure 4: Credit ratings for students and non-students

To test this difference, we conducted another two-sample t-test for difference in means with the following hypotheses.

$$H_0 : \mu_{\text{non-student}} = \mu_{\text{student}} \quad H_A : \mu_{\text{non-student}} > \mu_{\text{student}}$$

From this test, we generated a p-value of $p = 0.4879$. Since this value is significantly higher than our significance level of $\alpha = 0.05$, we will fail to reject our null hypothesis and do not have convincing evidence for non-students' average credit ratings to be higher than that of students.

Do married individuals have higher credit limits than unmarried individuals?

By looking at simply the distributions of credit scores for married and unmarried individuals, it appears that there may not be much of a difference (shown in Figure 5). However, we still wanted to test this to determine whether there actually was a notable difference.

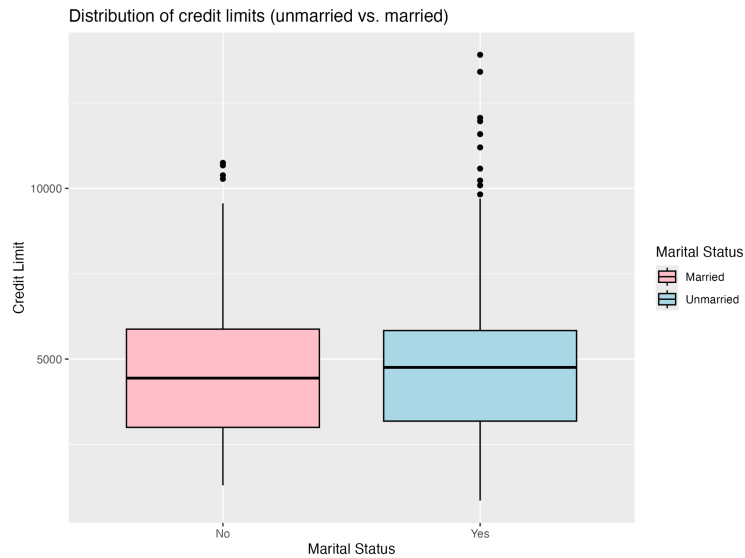


Figure 5: Credit Limits for Married and Unmarried Individuals

After creating a bootstrap distribution of differences in means between married and unmarried people (shown in Figure 6 below), where:

$$H_0 : \mu_{\text{married}} = \mu_{\text{unmarried}} \quad H_A : \mu_{\text{married}} > \mu_{\text{unmarried}}$$

We calculated a 95% confidence interval of this difference in means, with our resulting interval going from -304.1433 to 531.4101. Since 0 is contained in our interval, this means that 0 is a plausible value for the difference in mean credit limits, and thus we do not have convincing evidence that married people have a higher credit limit than unmarried people.

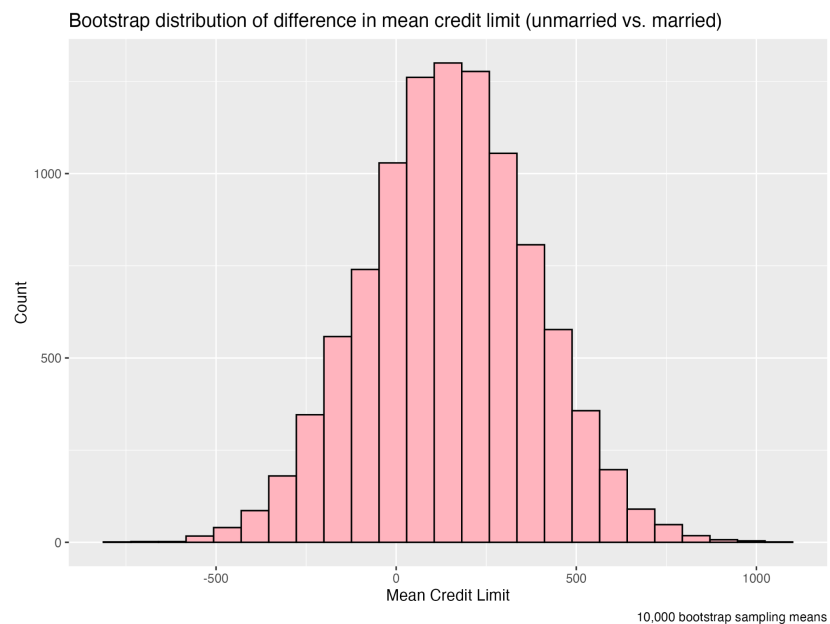


Figure 6: Bootstrap Distribution of the Difference in Mean Credit Limit, Married vs. Unmarried

Is there a relationship between marital status and credit score?

Finally, to test for a relationship between these two variables, we first plotted the distribution of credit scores for married and unmarried people (shown in Figure 7 below). We then conducted a Chi-square test for independence to determine if knowing a person's marital status has an effect on their credit score.

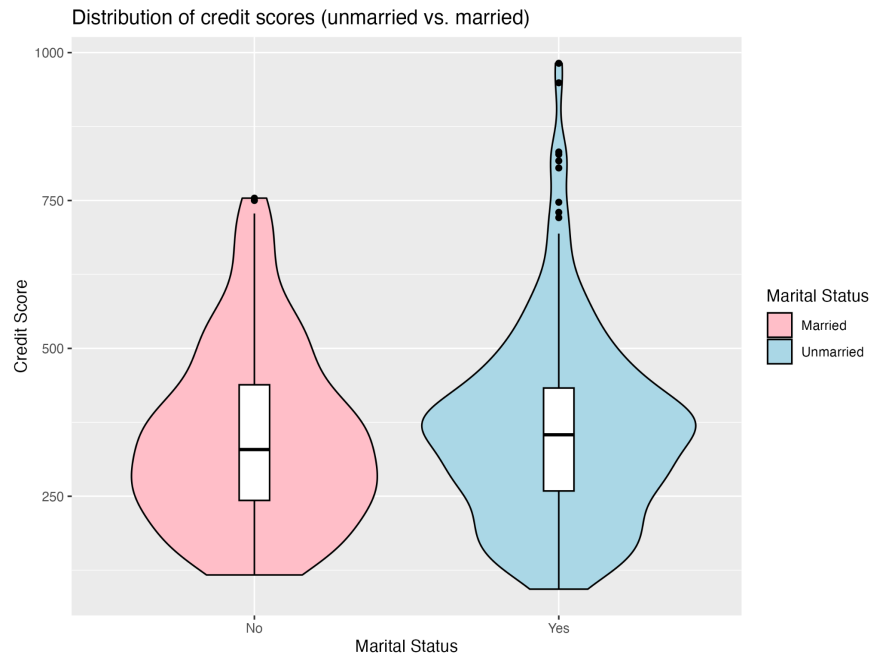


Figure 7: Credit Scores for Married and Unmarried Individuals

Our hypotheses for this test were as follows.

H_0 : There is no relationship between a person's marriage status and credit score

H_A : There is a relationship between a person's marriage status and credit score

After conducting our Chi-square test, we obtained a χ^2 value of $\chi^2 = 288.21$ and a p-value of $p = 0.3869$, with 2 degrees of freedom. Since our p-value is higher than the significance level of $\alpha = 0.05$, we fail to reject the null hypothesis and conclude that there is no significant evidence of a relationship between a person's marital status and their credit score.

Conclusion

Summary of Results

After completing our calculations, we found that credit rating and income had a significant impact on credit limit which confirmed our original inferences. We found convincing evidence that students had higher credit utilizations than non-students, but we found no significant difference in mean credit rating between students and non-students. Lastly, there was not significant evidence to support that marriage status had an impact on credit score. This contradicts our original hypothesis.

Critiques, Limitations, Validity of Data

Our presentation was limited by our knowledge and proficiency with both statistics and the use of the programming language R. This is apparent in the fact that we used very introductory level statistical methods that were easy to compute in R. It is expected we could have investigated more complex questions or employed different methods for our research questions with more learning of statistics. However, we believe that the analysis that was done was appropriate for our research questions and for the project.

Additionally, our data set was of course limited in the variables they provided. Further, the origins and description of variables of our data set was somewhat limited. For example, the education variable did not seem succinct enough to use as we were unsure whether this was in relation to college, trade school, or simply years in school. Also, it was not indicative of whether an individual passed this "level" of school or simply completed it. Though, the data set was large enough, allowing us to employ the statistical tests we did.

Future Questions

Some future analysis that we may want to perform regarding credit card would answer the following questions:

- How does debt levels affect credit score?
- What is the relationship between credit score and available mortgage rates?
- What is the relationship between credit score and available loan amounts?

Although uncertain of the methods with this specific data set, we think it would be interesting to interrogate how upbringing and education level impacts financial literacy and how that then translates to factors such as credit score and debt.

References

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